

Project Title:

Iram II Elementary School
Student Transcript and Academic Records System (STARS)

Role:

Lead Developer

Responsibilities:

- 1. Led the backend development of the system, including database design, API architecture, and server-side logic.
- 2. Integrated frontend and backend components to ensure seamless communication and data exchange.
- 3. Worked closely with frontend developers to align API responses with UI/UX requirements.
- 4. Ensured security, data integrity, and performance of the backend system.
- 5. Monitored progress and task completion across the development timeline.

Tasks & Contributions:

Task/Feature Worked on	Description	Status
Backend Architecture Design	Designed and implemented the system's backend infrastructure, including models and controllers.	Done
User Authentication & Session Control	Implemented secure login and session management features to control user access.	Done
SF10 Editing Module Development	Built core backend logic for creating, editing and updating student SF10 records.	Done
Student Record CRUD	Implemented Create function for student academic data.	Done
API Creation	Developed RESTful APIs to manage and serve student data and SF10 data.	Done

Error Handling & User Feedback	Integrated server-side validation messages and error responses to guide frontend data entry.	Done
Database Schema Design	Designed structured, normalized tables for storing detailed student and sf10 data.	Done

Challenges Encountered

- Complex Data Handling Requirements:
 - Implementing intricate logic to manage and process the extensive SF10 dataset has proven technically demanding, requiring thorough planning and testing to ensure accuracy and performance. Limited availability of the team due to overlapping academic and OJT responsibilities.
- Resource Constraints:
 - The team's availability is limited due to overlapping academic schedules and On-the-Job Training (OJT) responsibilities, impacting overall productivity and timelines.
- Unclear Project Specifications:
 - Some aspects of the project, particularly those related to beneficiary requirements, remain ambiguous. This uncertainty affects design decisions and may lead to rework once specifications are finalized.

Learnings and Reflections

Serving as the Lead Developer for the STARS project gave me a deeper appreciation for the responsibilities that come with technical leadership. Beyond writing code, I was accountable for the structure, reliability, and performance of the system's backend—especially the student records (SF10) module, which served as the core of the platform.

This role taught me how to coordinate development tasks, guide implementation standards, and help resolve technical blockers for the team. I had to ensure that our APIs were consistent, well-documented, and aligned with the needs of both the frontend

developers and the end users. It pushed me to think not just about how to solve a problem, but how to build solutions that others could depend on and build upon.

From a personal perspective, I grew significantly in backend development—particularly in data validation, API design, and database architecture. I learned how to manage complexity while keeping the system maintainable, and how to communicate technical decisions clearly to both technical and non-technical stakeholders.

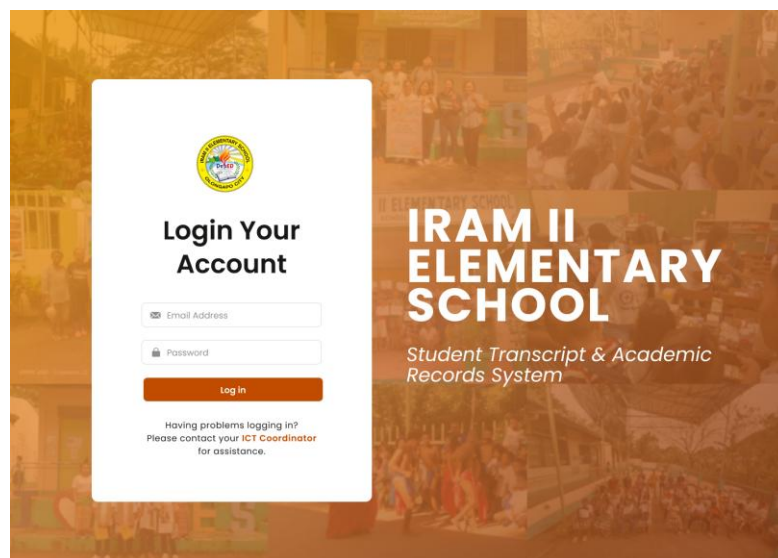
Overall, the project helped me grow as both a developer and a team leader, and showed me how backend systems play a crucial role in delivering real-world impact, especially in a setting like a school where accurate records matter.

Blog Link

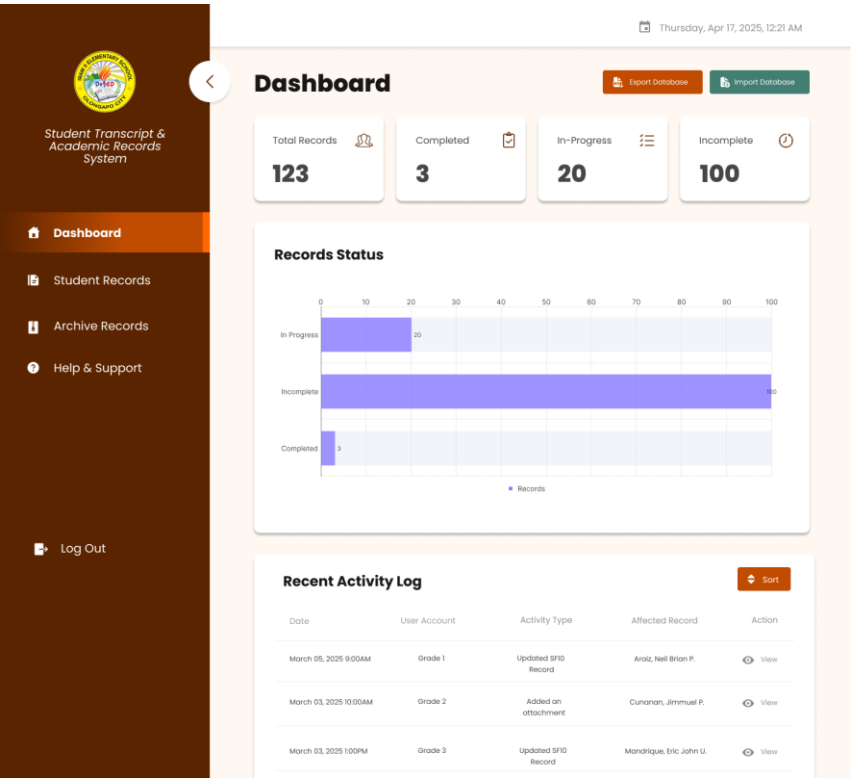
<https://blog.riandrei.tech/iram-ii-system>

Sample Work / Photo Documentation

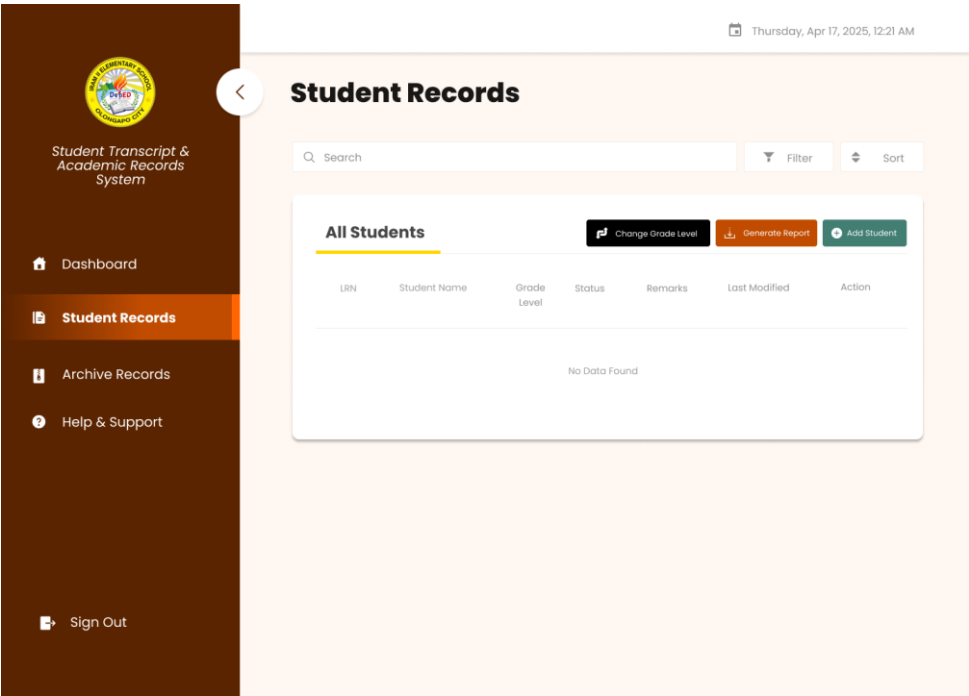
- Login Page



- Dashboard



- Student Records Tab





Student Transcript & Academic Records System

Dashboard

Student Records

Archive Records

Help & Support

Sign Out

Thursday, Apr 17, 2025, 12:21 AM

Student Records

Filter

Sort

All Students

Change Grade LevelGenerate ReportAdd Student

☐	LRN	Student Name	Grade Level	Status	Remarks	Last Modified	Action
☐	12345678	Aralz, Neil Brian P.	1	Enrolled	In-Progress	March 05, 2025 9:15 AM	ViewArchive
☐	12345678	Cunanan, Jimmuel P.		Enrolled Transferred In Transferred Out Non-Active Graduated	Completed	March 05, 2025 9:15 AM	ViewArchive
☐	12345678	Mandrique, Eric John U.	1	Enrolled	Completed	March 05, 2025 9:15 AM	ViewArchive
☐	12345678	Labuocca, Mark Christian G.	1	Enrolled	Completed	March 05, 2025 9:15 AM	ViewArchive

1

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3

4

• Others

Last Name

First Name

Name Extn. (Jr., I, II)

Middle Name



Republic of the Philippines
Department of Education

Learner Permanent Record for Elementary School (SF10-ES)
(Formerly Form 137)

LEARNER'S PERSONAL INFORMATION

LAST NAME: FIRST NAME: NAME EXTN. (Jr, I, II): MIDDLE NAME: Sex:

LEARNER REFERENCE NUMBER (LRN): Birthdate (mm/dd/yyyy):

ELIGIBILITY FOR ELEMENTARY SCHOOL ENROLLMENT

Credential Presented for Grade 1: ☐ Kinder Progress Report ☐ ECCD Checklist ☐ Kindergarten Certificate of Completion

Name of School: School ID: Address of School:

Other Credential Presented: ☐ PEPT Passer Rating: Date of Examination/Assessment (mm/dd/yyyy): ☐ Others (Pls. Specify): Remark:

PEPT Passer Rating: Name and Address of Testing Center:

SCHOLASTIC RECORD

School: Division: School ID: District: Division: School ID: Classified as Grade: Section: School Year: Region: Region:

Name of Adviser/Teacher: Signature: Name of Adviser/Teacher: Signature:

LEARNING AREAS

	Quarterly Rating	Final Rating	Remarks		
	1	2	3	4	
Mother Tongue					
Filipino					
English					
Mathematics					
Science					
Araling Panlipunan					
EPP/TLE					
MAPEH					
Music					
Art					

Learning Areas

	Quarterly Rating	Final Rating	Remarks		
	1	2	3	4	
Mother Tongue					
Filipino					
English					
Mathematics					
Science					
Araling Panlipunan					
EPP/TLE					
MAPEH					
Music					
Art					

Upload and Attach File

Attachments will be a part of this record.

PSA

Upload File

ECCD

Previous SF10

Other Document

Edit Document

Download PDF

Mark as Complete

Created: Today

Created by: Coordinator

Last Modified: 12/1/2025

Student Transcript & Academic Records System

Dashboard

Student Records

Archive Records

Help & Support

Sign Out

Thursday, Apr 17, 2025, 12:29 AM

Student Records

SearchFilterSort

All StudentsChange Grade LevelGenerate ReportAdd Student

Action Cannot Be Completed

Please ensure that all selected students are in the same grade level before proceeding.

Go Back

	Remarks	Last Modified	Action
	In-Progress	March 05, 2025 9:15 AM	View Archive
	Completed	March 05, 2025 9:15 AM	View Archive
12345678Mandique, Eric John U.	Enrolled	CompletedMarch 05, 2025 9:15 AM	View Archive
12345678Lalwica, Mark Christian G.	Enrolled	CompletedMarch 05, 2025 9:15 AM	View Archive

12345678

Student Transcript & Academic Records System

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Log Out

Thursday, Apr 17, 2025, 12:29 AM

Dashboard

Export DatabaseImport Database

Total Records0

Completed0

In-Progress0

Incomplete0

Records Status

Are you sure you want to import the database?

This action will override existing records. Ensure you have a backup before proceeding.

CancelConfirm Import

Recent Activity Log

Sort

Date	Name	Activity Type	Affected Record	Action
No Data Found				



Student Transcript & Academic Records System

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Sign Out

Thursday, Apr 17, 2025, 12:21 AM

Student Records

Add Student Record

Please complete all required fields to add a student record.

Last Name

First Name

Name Extension (Jr., I, II)

Middle Name

Grade Level

Select Grade Level

Cancel

Next



Student Transcript & Academic Records System

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Thursday, Apr 17, 2025, 12:21 AM

Student Records

Search

Filter

Sort

Confirm Changes in Grade Level ⓘ

Only the following students will have their grade levels updated:

Cunanan, Jimmuel P. (LRN: 12345678)

Mandrique, Eric John U. (LRN: 12345678)

Labucca, Mark Christian G. (LRN: 12345678)

Grade Level

Select Grade Level

Cancel

Confirm

☒	12345678	Labucca, Mark Christian G.	Grade 1	Enrolled	In-Progress	Scholastic Record	March 05, 2025 9:15 AM	View Archive
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